

ARMAVIR, Russian Federation

WMO: 370310

Lat: 44.983N

Lon: 41.117E

Elev: 159

StdP: 99.43

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-11.3	-17.7	0.8	-14.1	-14.7	1.1	-10.4	10.4	1.3	9.7	0.8	2.0	80	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	34.9	20.9	33.1	20.5	31.2	20.1	23.2	30.4	22.2	29.3	21.5	28.3	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	15.9	26.7	20.1	15.1	25.2	19.3	14.3	24.5	69.7	30.4	66.2	29.0	63.3	28.5	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	7.5	6.3	-19.0	37.2	4.2	1.9	-22.0	38.6	-24.5	39.7	-26.8	40.7	-29.9	42.1
			-18.9	24.4	4.3	1.4	-22.0	25.4	-24.6	26.2	-27.0	27.0	-30.1	28.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	-0.5	1.1	5.8	11.8	17.0	21.1	24.1	23.7	18.4	12.1	5.9	2.0	(6)
(7)		DBStd	9.57	5.30	5.64	4.23	4.12	3.82	3.00	2.72	3.12	3.81	4.65	4.72	4.67	(7)
(8)		HDD10.0	1161	326	251	140	27	2	0	0	0	1	29	136	249	(8)
(9)		HDD18.3	2859	584	483	387	198	72	9	1	2	44	198	373	507	(9)
(10)		CDD10.0	1870	0	1	11	81	217	334	438	424	254	96	13	1	(10)
(11)		CDD18.3	526	0	0	0	2	29	93	180	167	48	7	0	0	(11)
(12)		CDH23.3	5278	0	0	2	34	295	812	1872	1786	420	57	1	0	(12)
(13)		CDH26.7	2148	0	0	0	5	71	277	832	832	118	14	0	0	(13)
(14)	Wind	WSAvg	2.5	2.7	2.9	3.3	2.8	2.2	2.1	2.0	2.1	2.2	2.3	2.6	2.7	(14)
(15)	Precipitation	PrecAvg	572	34	30	37	53	66	72	56	57	42	41	43	40	(15)
(16)		PrecMax	836	70	82	74	156	122	205	139	138	86	88	89	110	(16)
(17)		PrecMin	387	8	1	0	6	6	3	11	9	17	6	5	7	(17)
(18)		PrecStd	104	16	20	19	35	28	47	32	40	21	25	24	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	16.6	21.2	27.0	30.8	34.0	36.7	37.3	32.2	28.2	20.3	13.7	(19)
(20)		MCWB	7.0	9.3	11.6	15.6	19.4	21.0	21.7	20.2	20.3	17.2	12.1	8.7	(20)	
(21)		2%	DB	8.5	12.7	17.9	23.8	28.8	31.8	34.9	35.3	29.8	24.5	16.8	10.8	(21)
(22)		MCWB	5.6	7.7	10.0	14.8	18.5	20.5	21.2	20.8	18.4	16.3	11.1	7.3	(22)	
(23)		5%	DB	6.7	9.9	15.0	21.5	26.7	29.8	33.2	33.4	27.8	21.6	14.2	8.9	(23)
(24)		MCWB	4.6	6.0	9.1	13.6	17.7	19.9	20.6	20.4	18.0	15.0	10.1	6.5	(24)	
(25)		10%	DB	5.1	7.6	12.3	19.1	24.6	27.8	31.2	31.3	25.7	18.8	12.0	7.3	(25)
(26)		MCWB	3.2	4.7	7.8	12.5	17.0	19.3	20.4	20.0	17.2	13.9	8.9	5.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	9.8	12.2	17.2	21.1	23.4	24.9	24.0	22.0	18.1	13.2	9.2	(27)
(28)		MADB	10.4	15.1	19.2	24.0	28.8	30.1	33.1	31.8	29.8	25.6	17.5	12.7	(28)	
(29)		2%	WB	5.9	8.0	10.7	15.4	19.6	22.0	23.4	23.0	20.2	16.8	11.8	7.8	(29)
(30)		MADB	8.1	12.4	16.5	22.1	26.5	28.9	30.4	30.7	26.6	23.0	16.0	10.3	(30)	
(31)		5%	WB	4.6	6.3	9.5	14.2	18.6	21.1	22.4	22.1	19.2	15.6	10.4	6.7	(31)
(32)		MADB	6.6	9.6	14.1	19.9	24.8	27.5	29.1	29.4	25.1	20.6	14.0	8.8	(32)	
(33)		10%	WB	3.4	4.9	8.1	13.1	17.6	20.2	21.6	21.3	18.2	14.4	9.0	5.5	(33)
(34)		MADB	5.0	7.4	11.9	18.1	23.0	25.9	28.2	28.1	23.8	18.3	11.6	7.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	6.6	8.0	8.7	10.3	10.5	10.6	11.9	12.5	11.3	9.5	8.2	6.5	(35)	
(36)		5% DB	MCDBR	8.5	11.6	14.2	14.1	14.2	14.0	14.9	16.0	14.6	14.5	12.6	8.8	(36)
(37)		MCWBR	5.9	7.0	7.5	6.7	6.1	5.2	5.0	5.1	5.6	6.7	7.2	6.0	(37)	
(38)		5% WB	MCDBR	8.0	10.9	12.6	12.2	12.4	11.8	12.1	12.7	12.1	12.5	11.5	8.2	(38)
(39)		MCWBR	5.8	6.8	7.0	6.2	5.9	5.1	5.0	5.0	5.4	6.0	6.8	5.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.370	0.417	0.463	0.427	0.434	0.460	0.457	0.421	0.388	0.351	0.338	(40)	
(41)		taud	2.297	2.206	2.121	2.036	2.227	2.244	2.187	2.196	2.268	2.319	2.364	2.315	(41)	
(42)		Ebn at Noon	755	793	803	798	845	838	810	796	793	766	735	710	(42)	
(43)		Edh at Noon	83	109	135	160	137	135	142	135	116	96	77	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.40	2.29	3.14	4.64	5.70	6.23	6.35	5.71	4.23	2.69	1.63	1.17	(44)	
(45)		RadStd	0.19	0.32	0.28	0.47	0.63	0.38	0.31	0.33	0.46	0.33	0.19	0.13	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (9 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	-122	+118	N/A	

Nomenclature: See separate page